

Complete Digital Library Ecosystem

Project	Gyaana Grama —
Location	Sathnur Village Library, Karnataka, India
Platforms	Windows Desktop App + Android Mobile App
Team	YESH (Desktop) Paul (Android) Adithya (Field)
Status	Phase 1 Complete — Ready for Handover to Library
Goal	Free, offline-first, bilingual library management for rural India

Books Digitised

Village Members

Android Companion

Desktop System

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THE VILLAGE LIBRARY PROBLEM

Why Sathnur needed a digital solution

The Sathnur village library serves hundreds of students, farmers, and families in rural Karnataka. For years, every book borrowed, returned, or registered was tracked by hand — in paper registers that could be lost, damaged, or were simply impossible to search. Staff spent more time managing paper than helping readers find books.

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A student would ask if a specific book was available. The librarian would flip through pages of handwritten entries — sometimes for 10 minutes — before giving an answer.

— Field observation, Sathnur Library

■ Paper Registers	Books tracked in handwritten ledgers — no numbering, no index, no search.
■ No Search	Finding a book meant scanning every page. Students often gave up and left.
■ Language Gap	All existing library software was English-only. Staff and students read Kannada.
■ Costly Options	Commercial systems cost thousands per year and require constant internet.
■ No Digitization	Adding new books required 10 minutes of manual typing per book.
■ No Internet	Sathnur has intermittent connectivity — cloud tools were never realistic.

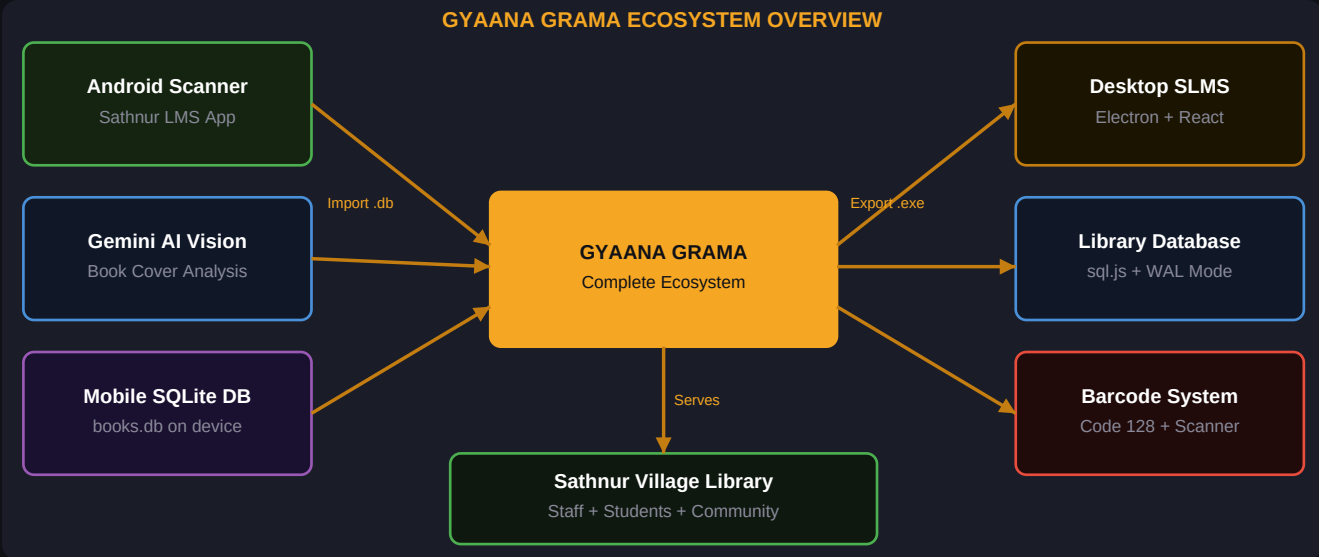
What was needed: completely free, no internet required, Kannada interface, usable by staff with no software training, running on the library's old Pentium computer with 4 GB RAM.

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THE COMPLETE SOLUTION

Two apps, one ecosystem — built for village life

Gyaana Grama is not a single application — it is a complete digital ecosystem. Two apps work as one: a mobile Android scanner that staff use to add books by photographing them, and a powerful desktop system where the full library is managed, searched, and operated every day.



The Gyaana Grama ecosystem: Android app feeds data into the Desktop SLMS, which serves the village library daily

■ Android LMS Scanner

Built for the librarian's phone. Staff walk through shelves, point the camera at a book cover, and AI instantly reads the title, author, and publisher. The book is saved in seconds. No typing, no mistakes, no wasted time. Flutter · Gemini AI · SQLite · Code 128

■ Desktop SLMS

The full library management system on the library's Windows computer. Staff issue and return books, search the catalogue instantly, track members, and view everything on a clear dashboard — all in Kannada or English. Electron · React · sql.js · Tailwind

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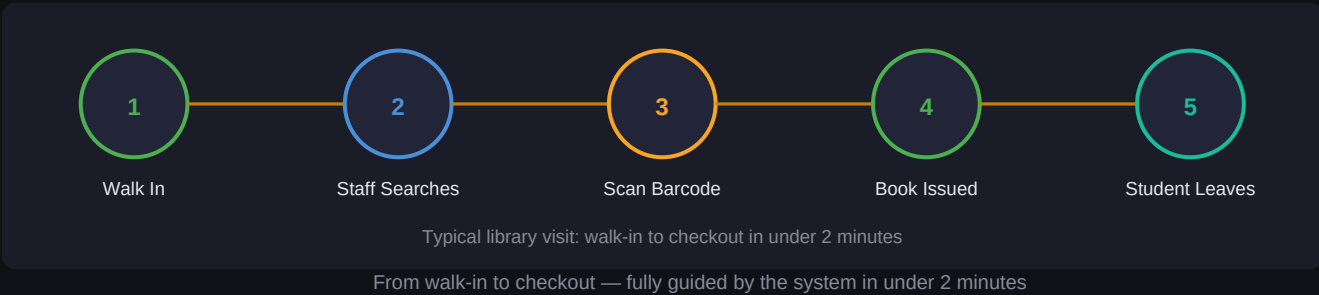
WHO DOES THIS HELP & HOW

Real benefits for real people in Sathnur

The Gyaana Grama ecosystem was designed with one question: how does this make life better for the actual people who use the library every day?

■ Students & Young Readers	Students can now ask for a book and get an instant answer. Search takes seconds, not minutes. Students see exactly when borrowed books are due, and their full borrow history is tracked. More time reading, less time waiting.
■ Library Staff & Librarians	Daily tasks of issuing and returning books are guided step-by-step on screen, in Kannada. Mistakes are caught before they happen — the system warns if no copies are available or if a due date is wrong.
■ Village Community Members	Community members get a permanent MEM-ID. Their borrowing history is preserved digitally. The library can now report how many books it has, which genres are popular, and how active its membership is.
■ The Library as an Institution	The library has a complete searchable catalogue with accession numbers, barcodes, genre tags, and language classification. It can grow its collection systematically using the Android scanner and export data for official records.
■ The Village Education Ecosystem	A well-run library encourages reading habits and academic curiosity. By removing the friction of manual management, staff can focus on connecting readers with books — not managing paper.

A Typical Library Visit



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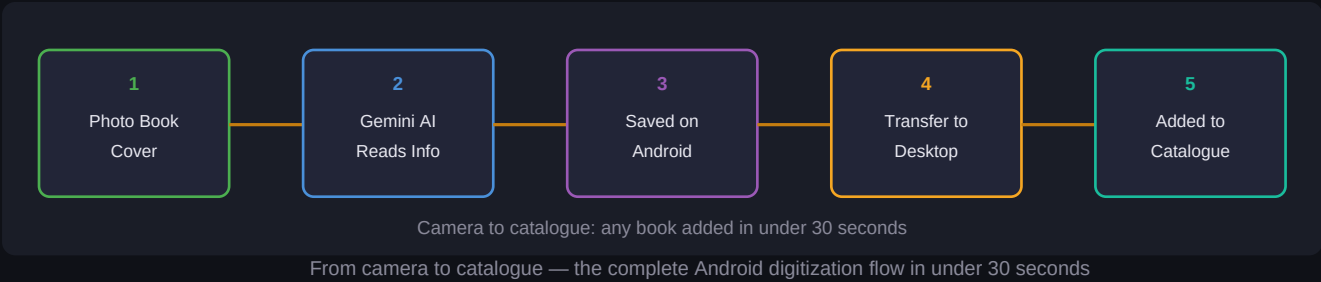
THE ANDROID APP — SATHNUR LMS SCANNER

Phone-based digitization and catalogue companion

The Sathnur LMS Scanner is a free Android application that turns any smartphone into a complete book-cataloguing workstation. It solves the hardest problem in rural library digitization: how do you add hundreds of books to a digital system without spending hours typing every detail?

The Answer: Point, Shoot, Done

Staff open the app, tap Scan, and point the phone camera at a book cover. Within seconds, Google Gemini AI reads the image and fills in the title, author, publisher, year, and genre automatically. The book is saved instantly. No typing. No internet needed for day-to-day use.



What the Android App Can Do

■ Camera Barcode Scanner	Points at any barcode — instantly reads the book’s accession ID and looks up its record.
■ AI Cover Analysis	Photograph the book cover; Gemini AI extracts title, author, publisher, year, genre automatically.
■ Manual Entry Fallback	If camera is unavailable, staff can type barcode or accession number directly.
■ 3-Language UI	Full interface in English, Hindi, and Kannada — switchable from Settings instantly.
■ Real-Time Search	Search the full catalogue by title, author, accession ID, or ISBN instantly.
■ PDF & CSV Export	Export catalogue as PDF barcode labels or a spreadsheet — one tap.
■ Database Backup	Export the full database to USB or Google Drive for safekeeping.
■ Bulk CSV Import	Import books in bulk from a spreadsheet — for libraries with existing records.
■ Soft Delete + Undo	Deleted books go to Trash. Recoverable until permanently removed.
■ Configurable Settings	Library name, accession prefix, default language, label sizes — all adjustable.

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Without AI, adding 500 books to a digital system would take 80+ hours of manual typing. With Gemini AI cover scanning, the same task takes under 5 hours — one photo per book.

— Estimated based on field benchmarks

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THE DESKTOP APP — GYAANA GRAMA SLMS

The daily management hub for library staff

The Gyaana Grama SLMS is the heart of daily library operations. It runs on the library's existing Windows computer — no installation needed, no internet required — and gives staff a clear, modern interface to manage every aspect of the library in Kannada or English.

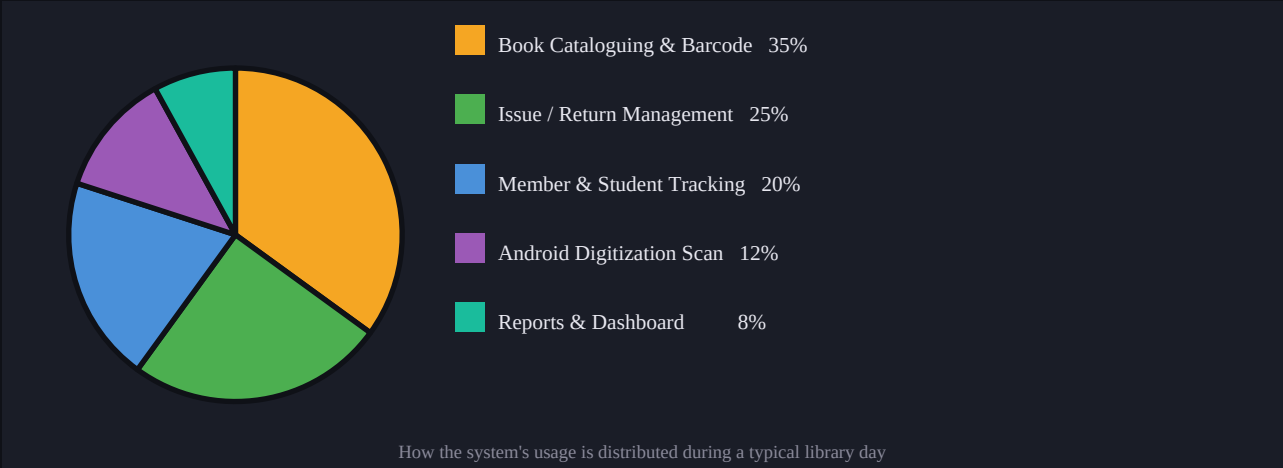
What Staff See and Use Every Day

■ Dashboard	Opening screen: total books, members, active loans, overdue books. Staff know the library's status before the first patron walks in.
■ Book Catalogue	Complete searchable list of every book — by title, author, accession number, or barcode scan. Filter by genre, language, availability.
■ Member Registry	Every member has a MEM-XXXX ID. Staff see current loans and full borrow history on one screen.
■ Issue a Book	Scan the book barcode, select the member, confirm the due date — issued. System validates copies and due dates before allowing it.
■ Return a Book	Scan the returning book. System shows who borrowed it and when. Staff confirm and the return logs instantly.
■ Settings	Library name in Kannada and English. Configurable loan period. One-tap language switch.

Built for Staff Who Have Never Used Library Software Before

- ◆ Every button and label appears in the chosen language — Kannada or English — at all times.
- ◆ Action buttons are always visible. Nothing hidden behind hover effects or collapsed menus.
- ◆ Confirmation steps protect against mistakes: deleting or returning requires an explicit confirm.
- ◆ The Retsol LS-450 barcode scanner plugs into USB and works like a keyboard — scan and go.
- ◆ All transitions are fast on the library's old Pentium computer — tuned for low-spec hardware.
- ◆ No internet ever required. Completely self-contained in a single portable file.

How daily usage is distributed across system features



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HOW THEY WORK TOGETHER

The integration that makes the ecosystem complete

The Android app and Desktop SLMS are designed to complement each other perfectly. The Android app excels at rapidly adding new books — camera in hand, walking the shelves. The Desktop SLMS excels at daily operations — issuing, returning, searching, managing members. Together they cover the complete lifecycle of every book in the library.

The Transfer Process — Simple and Reliable

1

Add Books on Android

Staff photograph and register new books during a cataloguing session.

2

Export from Android

Entire database exported as a .db file to Google Drive or USB with one tap.

3

Import on Desktop

Open SLMS, tap "Import from Android", pick the file from the file browser.

4

Automatic Merge

New books get GG-XXXX accession numbers. Duplicate books detected automatically.

5

Ready to Use

All newly registered books appear in catalogue immediately — searchable and issuable.

Same Barcode, Both Systems

The Code 128 barcode generated by the Android app is the same barcode read by the desktop scanner. A book registered on the phone as GG-0247 can be immediately scanned at the desk to issue it. The unified GG-XXXX accession format means there is never any confusion about which book is which.

Feature Coverage Across Both Platforms

	Book Catalogue	Barcodes	Issue/Return	Members	Kannada UI	AI Scanning	Offline	Export	Search
Desktop	✓	✓	✓	✓	✓	—	✓	—	✓
Android	✓	✓	—	—	✓	✓	✓	✓	✓

Feature coverage: together the two apps cover the complete library workflow

Green = Desktop | Amber = Android | Together they deliver complete library management

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FULL FEATURE COMPARISON

Everything the ecosystem delivers

Feature	Android App	Desktop SLMS	Benefit to Library
Book Registration	✓ Camera + AI	✓ Form + Scanner	Any book added in under 30 seconds
Barcode Generation	✓ Code 128 PDF	✓ Code 128 Print	Scannable labels for every book
Real-Time Search	✓ By all fields	✓ By all fields	Find any book instantly
Issue & Return	—	✓ Full workflow	Step-by-step guided with validation
Member Management	—	✓ MEM-XXXX	Full member registry with history
Kannada Language	✓ Full UI	✓ Full UI	Staff can work entirely in Kannada
Hindi Language	✓ Full UI	—	Accessible to Hindi-speaking users
Dashboard & Stats	—	✓ Summary cards	Instant library health overview
AI Cover Scanning	✓ Gemini AI	—	Photograph = auto-fill all metadata
CSV Export	✓ Configurable	—	Spreadsheet-ready catalogue export
Database Backup	✓ One tap	—	Full backup to USB or Drive
Soft Delete + Undo	✓ Trash view	✓ Confirm dialog	No accidental permanent losses
Bulk Book Edit	✓ Multi-select	—	Edit genre or language for many books
Offline Operation	✓ 100%	✓ 100%	Zero internet dependency ever
Portable / No Install	—	✓ Single .exe	Runs from USB, no setup required

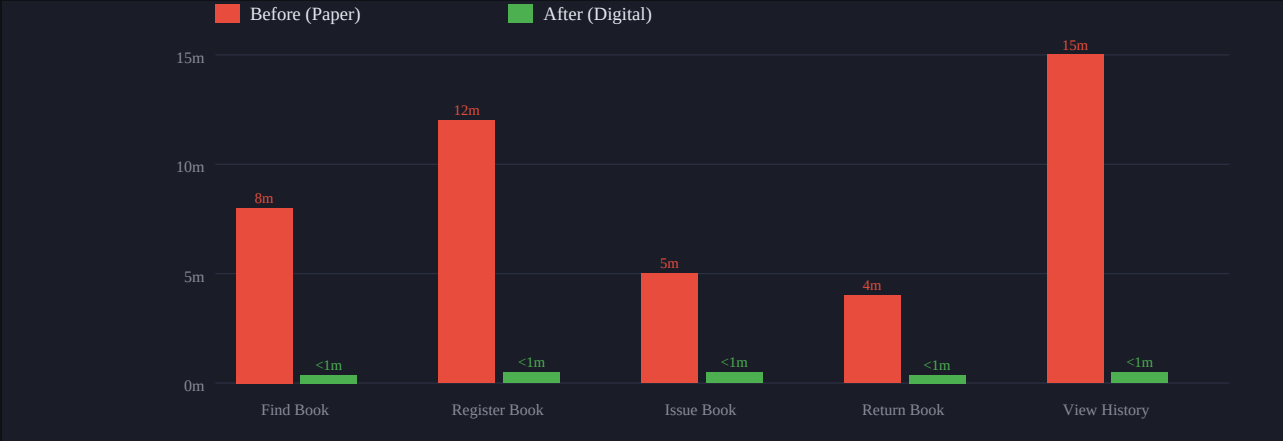
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REAL-WORLD IMPACT & TIME SAVED

What changes when Gyaana Grama is running

The numbers tell part of the story — but the real impact is in everyday moments that are no longer frustrating. A student gets an answer in 10 seconds instead of 10 minutes. A staff member issues 20 books in the time it used to take to issue 3. A librarian finally knows exactly how many books the library owns.

Time Saved on Common Tasks



Average minutes per task — Paper system (red) vs. Gyaana Grama digital system (green)

Concrete Before vs After

Before the System	After Gyaana Grama
10 min to check if a book is available	Under 10 seconds with instant search
5–10 min to type in a new book manually	30 seconds with AI camera scan
No way to know who has borrowed what	Full borrow history per member
English-only software options	Complete Kannada interface
Paper registers could be lost or damaged	Backed-up digital database file
No barcode system at all	GG-XXXX Code 128 labels on every book
No permanent membership record	MEM-XXXX for every library member
No visibility into overdue books	Dashboard shows overdue books at a glance
Expensive or unavailable software	Completely free, forever

Feature Completion

Book Catalogue (CRUD + Barcode)		100%
Member Management		100%
Issue & Return Flow		100%
Dashboard & Analytics		100%

Bilingual UI (EN + Kannada)		100%
Android Import Integration		100%
Barcode Scanner Support		100%
User Guides (EN + Kannada PPTX)		100%
Portable .exe Build		100%



DESIGN PRINCIPLES & DEVELOPMENT STORY

How the team thought about building this

Every decision — from the colour of the interface to the database engine — was made with one question: will this work reliably for the Sathnur library? Not for a tech company’s servers or a developer’s high-end laptop — for a Pentium computer in a village library, used by staff who have never used library software before.

Guiding Principles

■ Offline First, Always

No feature requires internet to work. The Android app runs without WiFi. The desktop app runs from a single file with no cloud dependency. This was a hard requirement from day one.

■ Kannada is Not Optional

Both apps offer complete interfaces in Kannada — every label, error message, and button. The desktop can switch to Kannada with one tap and stays that way across restarts.

■ Fast on Old Hardware

The desktop app was tuned for the Pentium G630 with 4 GB RAM. Zero GPU-heavy animations, instant button responses, and a database engine that needs no installation.

■ Always Visible, Never Hidden

Staff should never hunt for a button. Every action is permanently visible — nothing hidden behind hover effects or tiny icons.

■ Mistakes are Caught Early

The system validates before acting. Issue a book with no copies? Blocked. Due date in the past? Blocked. Returning the wrong book? A screen shows who borrowed it before confirming.

■ Simple to Deploy

The desktop app is a single portable .exe file. No installation, no admin rights. Place on Desktop, double-click, and it works. Database is created automatically on first launch.

Key Challenges the Team Solved

Database without Installation	Standard SQLite libraries require compiling native code — which fails on older setups. Replaced with sql.js, a pure-JavaScript SQLite engine needing zero compilation and working on any Windows machine.
AI Scanning without Internet	Gemini AI requires internet for cover scanning — but day-to-day library operations do not. AI scanning is an optional enhancement; the system works fully offline without it.
Error Messages in Kannada	Early in development, all error messages showed as generic text because the translation system was missing a placeholder for the actual error content. Once fixed, staff could read exactly what went wrong — in Kannada.
Speed on Low-End Hardware	Animations, backdrop-blur effects, and heavy transitions cause visible lag on Pentium-class CPUs. All were removed or capped at 200ms. The app responds as fast on the library machine as on a modern laptop.

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THE TEAM BEHIND IT

Three people, one shared goal

Gyaana Grama was built by a small team with a shared commitment to making technology work for rural India — not the other way around.

YESH

Desktop Lead Developer

Designed and built the entire Gyaana Grama SLMS desktop application — architecture, database, user interface, bilingual support, barcode system, build pipeline, Android integration, and all project documentation. Coordinated the integration between the desktop and Android apps from concept to handover.

Paul

Android Developer

Built the Sathnur LMS Scanner — the Flutter-based Android companion app. Implemented Gemini AI cover scanning, barcode generation, SQLite mobile database, multi-language support, CSV/PDF export, and the database transfer mechanism used to sync with the desktop.

Adithya

Hardware & Field Research

Validated the system on the actual library hardware — the Pentium G630 computer at Sathnur. Conducted field research to understand staff workflows, identify friction points, and confirm that the system performs in real village conditions.

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This project proves that the tools for modernising rural institutions already exist — they just need to be assembled thoughtfully, with the real users in mind from the very first day.

— Project Retrospective

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Handover & What Comes Next

Delivering the system to Sathnur library

Phase 1 of Gyaana Grama is complete. Both apps are stable, tested, and ready for the library team to use. The handover is designed to be as simple as the software itself.

What the Library Receives

■ Desktop App (.exe)	Single portable file. Place on Desktop, double-click to launch. No installation.
■ Android App (.apk)	Install on library staff's Android phone. Works immediately, no account needed.
■ User Guide — English	13-slide presentation covering every feature with step-by-step instructions.
■ User Guide — Kannada	13-slide guide fully translated into Kannada — primary reference for library staff.
■ Barcode Scanner	Retsol LS-450 USB scanner pre-configured and ready to use with the desktop.
■ Pre-loaded Settings	"Gyaana Grama Library" and Kannada library name pre-configured in the system.
■ Hardware Setup	Adithya confirms the Pentium G630 machine is configured with the .exe on the Desktop.

Staff's First Week — Getting Started

- 1

Launch the Desktop App

Double-click the .exe. Database creates itself automatically on first launch.
- 2

Switch to Kannada

Settings → Language → . Full UI switches in one tap.
- 3

Register Members

Add library members using the Member form. MEM-IDs are assigned automatically.
- 4

Scan Books with Android

Use Paul's app to photograph and register the book collection shelf by shelf.
- 5

Import to Desktop

Transfer the Android database and import — all books appear instantly.
- 6

Start Issuing Books

Scan a barcode, select a member, confirm. The library is now fully digital.

Looking Ahead — Possible Future Features

Phase 1 focused on what the library needs today. The following are documented as possibilities for a future phase, to be built only if the library grows into needing them.

■ Fine Calculation	Auto-compute overdue fines based on a daily rate — if the library decides to implement them.
■ Reservation Queue	Allow members to reserve a book currently on loan. Queue management.
■ Monthly Reports	Auto-generated PDF reports: most-borrowed books, active members, genre popularity.
■ Multi-Library Network	If the Sathnur model expands to neighboring villages — a shared regional catalogue.

GYAANA GRAMA — PHASE 1 COMPLETE

Desktop SLMS + Android LMS Scanner | Sathnur Village Library, Karnataka
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